

Planning Fast and Slow: Guided Trajectory Synthesis via Conditional Flow Matching

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ABSTRACT

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Note that this is the first thing that the examiner is going to read possibly after taking a glance at the title. They need to be informed about what they are going to expect to find in this report.

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1 INTRODUCTION

Imbuing intelligent agents with the capability for autonomous-decision-making in real-world environments is challenging. For decades, reinforcement learning has offered a robust mathematical framework for acquiring optimal behavioral skills through interactions within an environment to maximize cumulative rewards (Sutton and Barto 2014). The integration of deep neural networks as high-capacity function approximators has propelled RL even farther, leading to remarkable achievements across diverse domains, from mastering complex games to enabling basic robotic locomotion and manipulation (Chen et al. 2018).

However, the widespread success of modern machine learning, as seen in areas such as computer vision and natural language processing, fundamentally hinges on the availability of large and highly diverse datasets. In the conventional RL paradigm, the acquisition of such extensive data becomes prohibitively impractical for real-world applications. For example, autonomous driving would need to practice in numerous cities for every model update, which is costly, time-consuming, let alone being unsafe. This inherent limitation created a gulf of generalization between lab-bound RL successes and the complex, open-world settings where intelligent decision-making is most critically needed.

Offline Reinforcement Learning emerges as a pivotal paradigm to bridge this gap. Offline RL focuses on enabling the training of policies solely from previously collected datasets, without additional online interaction. This approach holds immense potential for transforming vast amounts of historical data into robust, generalizable decision-making engines, much like how supervised learning revolutionized pattern recognition. The implications for fields such as robotics, healthcare, and autonomous driving are profound, as it offers a viable pathway to deploy sophisticated AI systems where active data collection is either unfeasible or too costly. Beyond these, offline RL can revolutionize domains such as operations research, logistics, power plant control, inventory management, and finance, all of which have rich historical data where active exploration is difficult or expensive (Levine et al. 2020).

Finally, taking a page out of the rise of vision language models, the ability to leverage large and diverse datasets in an offline setting further enables the development of foundational Vision-Language-Action models or generalist robot models. This means that pre-training on massive, heterogenous and potentially multi-modal datasets can lead to acquisition of broadly generalizable skills. These can then be efficiently fine-tuned for specific tasks in deployment. In a way, this paradigm allows bootstrapping on pre-trained visual language models to learn behaviors. Passive data can now be transformed into active intelligence by learned observations (Kim et al 2024).

1.1 Background and Context

Despite its immense promise, offline RL is plagued by algorithmic challenges. At its core, offline RL is fundamentally about making and answering counterfactual queries (Levine, 2020). These are “what if” questions about what might happen if the agent were to carry out actions different from those in the dataset. In other words, for an agent to improve upon the behavior policy, it must be able to execute action sequences that diverge from the training data. This inherent need for counterfactual inference strains conventional machine learning tools, which assume that test data is drawn from the same distribution as training data. Consequently, making such counterfactual queries results in distributional shift. When a learned policy deviates from the training data, conventional RL algorithms can produce erroneous and excessively optimistic Q-value estimates, leading to brittle policies that perform poorly when deployed.

This instability is particularly exacerbated by high-dimensional function approximators such as Deep Neural Networks (DNNs). The errors can accumulate over long horizons, making stable learning notoriously difficult (CITE HERE). Furthermore, real-world datasets, especially those from human demonstrations, inherently exhibit multimodal action distributions; there are often multiple valid ways to perform a task. Traditional methods like behavior cloning struggle to capture this multimodality, often averaging across modes and leading to indecisive or jerky behaviors that fail to generalize.

To address the challenge of multimodality, generative modeling, particularly with denoising diffusion models, has emerged as a powerful approach. By framing planning as a conditional sampling process, diffusion models can learn and represent complex, multimodal distributions directly from data. This is a critical capability for robotics, where a

task may have multiple valid solutions, such as different grasps for an object or various paths around an obstacle. By capturing the entire distribution of expert solutions, these models can enhance a robot’s versatility and ability to generalize.

However, the very properties that make diffusion models powerful for generation introduce fundamental challenges for their deployment in real-time robotic systems. These models are built upon a stochastic, iterative framework that imposes a significant computational burden, leading to high-latency inference incompatible with dynamic control loops. More importantly, their stochastic nature is fundamentally misaligned with the deterministic physical laws and hard safety constraints that govern robotic motion, calling for an alternative generative paradigms.

Therefore, this dissertation presents Flow Planner, a hierarchical framework built upon flow matching, a powerful alternative that directly addresses the core limitations of diffusion models. Flow matching learns a deterministic transformation governed by an Ordinary Differential Equation (ODE) rather than a Stochastic Differential Equation (SDE). This fundamental distinction makes it not merely an incremental improvement but a paradigm inherently aligned with the deterministic, low-latency requirements of physical robotic systems.

1.2 Objectives

The primary aim of this dissertation is to develop and validate a novel, deterministic framework for behavior cloning in offline reinforcement learning, leveraging the principles of flow matching. The specific objectives established to achieve this aim are as follows:

- To design and validate a systematic methodology for curating offline reinforcement learning datasets from actively trained agents.
- To design and implement "Flow Planner," a modular framework that utilizes flow matching for the generation of trajectories from offline data.
- To analyze the impact of various conditioning mechanisms on the precision and generalizability of the trajectories generated by the framework.

- To evaluate the performance of the "Flow Planner" framework, benchmarking its core attributes such as inference speed and policy effectiveness against diffusion models.

1.3 Achievements

This dissertation's primary contribution is the development and validation of Flow Planner, a novel and modular framework for deterministic behavior cloning. The key achievements are:

- **A Data Curation Pipeline and Benchmark Datasets:** A flexible pipeline was designed and implemented for curating offline reinforcement learning datasets. This was demonstrated by producing three distinct datasets from PPO agents in Box2D environments, with the system designed for easy extension to other agents and environments.
- **A Modular Framework for Behavior Cloning:** The Flow Planner framework was developed to directly address the high latency and stochasticity that limit current diffusion-based BC approaches. Its modular architecture is a key feature, allowing for the flexible optimization of its different components.
- **Analysis of the Flow Planner Framework:** This work provides a comprehensive analysis of the architectural choices essential for applying flow-matching models to behavior cloning. The findings highlight the critical role of conditioning mechanisms and model architecture, offering practical guidance for future implementations.
- **Successful Proof-of-Concept Validation:** The Flow Planner framework was successfully validated in the LunarLander continuous control environment. This proof-of-concept demonstrated the ability to generate effective policies with significantly lower inference latency than baseline approaches.

1.4 Overview of Dissertation

This dissertation is structured as follows:

- Chapter 2 provides the theoretical background for this work. It begins by reviewing foundational concepts in offline reinforcement learning and methods for addressing

distributional shift. The chapter then provides an overview of Denoising Diffusion Probabilistic Models, unifying these concepts by framing them through the lens of continuous flows to introduce flow matching. It concludes by reviewing applications of diffusion for trajectory synthesis and related state-of-the-art attempts at integrating these models directly into reinforcement learning.

- Chapter 3 details the methodology for data curation and introduces the experimental environments. This includes a technical discussion on the implementation of the data pipeline using Gymnasium wrappers and the Minari library for dataset management.
- Chapter 4 presents the core technical contribution of this dissertation. It details the different approaches to model the control problem and introduces the proposed hierarchical planning framework, "Flow Planner," along with its implementation details using Meta's flow matching library.
- Chapter 5 provides a comprehensive analysis and discussion of the empirical results. This includes an investigation of various conditioning mechanisms and a benchmark comparison of the framework's performance against baseline models.
- Chapter 6 concludes the dissertation by summarizing the key contributions, discussing the implications of the findings, and outlining promising directions for future research.

2 BACKGROUND THEORY AND LITERATURE REVIEW

This chapter is required to include a more deep literature review beyond what you briefly introduced in section 1.1. Furthermore it should include the background theory and concepts to what is critical to your work. It is very important to consider what sections you should form within this chapter and how to structure it. You should send a planned contents of this and other chapters to your supervisor before you write up and ensure they are happy with what you propose to do.

The earlier sections of this chapter should include a literature review of the state of the art in your topic area citing books, publications and possibly some other web sources of work relevant to the topic. They should be written up in a way that they lead the reader to realising where your contributions to the work are useful and filling a gap. Don't just assume the reader will know what you write about, as your second examiner is not considered to be expert in the area and will want to learn from what you write about.

There will be background theory and concepts that are directly relevant to your work. It is important to work out what these are and develop appropriate sections for this purpose. Again the reader may need to learn about these so it is important that you explain them clearly and in the context that is relevant to your work, not just repeating what all the other literature says. You need to show you understand it by explaining it.

You should ensure that at this point in the chapter a paragraph or two is presented to show what content will be in this chapter and this significance in order to indicate where it is going to lead the reader.

2.1 Section Title

Text for section

2.2 Section Title

Continue with as many sections as you need

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2.3 Summary

Your final section should include a summary of the chapter, which will summarise what has been covered and where there are gaps in knowledge which pave the way into the original contributions you make in the coming chapters.

3 TECHNICAL CHAPTER

The technical body of the report consists of a number of chapters (just one here, but there will usually be more) of work you have carried out to date. Follow a logical structure in how you present your work. This will usually be the phases of the work you carried out which will link loosely to the objectives. Note that these chapters are not meant to be a log book, thus do not write them like a diary of work carried out but as a technical report specifying clearly what has been implemented and presented in a way that someone else could carry out the work based on what they have read.

Again you need to consult with your supervisor on the technical chapters you plan to write by providing a draft contents page. It should show clearly the sections in it and every chapter should have a clear paragraph introducing it and what to expect in it.

3.1 First Section

Structure your text into sections.

3.1.1 First Subsection

If necessary, also use subsections. Subsections are entered using the Heading 3 paragraph style (all these heading styles are self-numbering).

3.1.1.1 First Subsubsection

If you really need subsubsections.

3.1.2 Second Subsection

And, as required, more subsections.

3.2 Second Section

As an example of a figure, consider Figure 3.1 and note how it is referenced here by cross referencing to a caption method so that also you can form a list of figures at the start. You should be familiar with how to insert a caption and refer to it in Word from the training needs analysis form you were given. The caption should include title and essential

information about figure and figures must always be referred to in the text. Ensure you insert figures into appropriate points in the text so they are both near to the text that is referring to the figure but also ensure that they are arranged in a tidy fashion so they do not leave big gaps with no text.

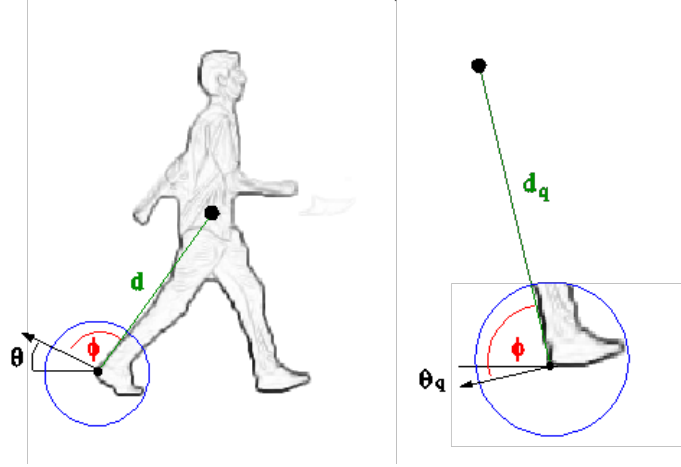


Figure 3.1: Example figure and caption.

Table 3.1: Make sure to set caption to large size.

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3.3 Third Section

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$$y = x^2 \tag{3.1}$$

but note it is usually after putting in an equation that you may refer back to equation (3.1) later in the document or chapter.

4 TECHNICAL CHAPTER 2

You need to work out how many technical chapters you will require, possibly more than two in this template. Normally your final chapter before the conclusion will include some clear results or proof of concepts from the end goals of your dissertation and it is important to document these well and show the value of what you have achieved. Frequently many dissertations fail to document this information correctly because it is rushed at the end so do ensure you allow time to write it properly.

4.1 Section

Text to insert

4.2 Section

Text to insert – but ensure the last section really does show the value of your work and where you have got to.

5 CONCLUSIONS

Your conclusion should summarise what you have carried out in this work to date, what you have achieved and what you are on course to achieve. Also it should explain what was documented in this dissertation. Note that someone may read this chapter without bothering to read the earlier chapters so there is no problem with repeating what you may have already written earlier. Indeed no new information should be in this part of the conclusion. It should reference back to what is already there.

5.1 Evaluation

Stand back and evaluate what you have achieved and how well you have met the objectives. Evaluate your achievements against your objectives in section 1.2. Demonstrate that you have tackled the project in a professional manner.

5.2 Future Work

Discuss what further work could be carried out from this work and relate it to what has been carried out.

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APPENDIX

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Note that you should not include your program listings as an appendix or appendices. You should submit one copy of such bulky text as a separate item, perhaps on a disk.